

General Industries Hub

Enterprise Inventory Management

Standard Operating Procedure - Procurement Chain

Version 3.0 - Generated 2026-10-03

Confidential - for authorized personnel only

GI Hub - Multi-Site Inventory, Procurement & Asset Management

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General Industries Hub - Standard Operating Procedure

Version 2.0 - 3 October 2026 - The whole system, by role Owners: Site HOD Council - Logistics Manager - Warehouse Lead - GI Hub Administrator Applies to: every site and warehouse on GI Hub (the React / FastAPI application) Companion documents: USER_MANUAL.md - every page, field and button (the § numbers below point into it) - MANUAL_TESTING_GUIDE.md - how each feature is checked - docs/ARCHITECTURE.md - how the system is built Review: quarterly, and whenever a role, a workbook or an approval step changes.

> What changed from v1.0. Version 1.0 (July 2026) covered the procurement > chain only, and was written for the old Streamlit screens. Version 2.0 covers > the whole system - stock entries, consumption forms, the Surface Shield queue, > lots and FEFO (Phase 16), Practice, the Excel workbook sync and the > administrator's routine - and is organised by role. The procurement decision > trees, escalation matrix and recovery procedures from v1.0 are kept in §6-§8, > unchanged in substance.

§1. Purpose & scope

This SOP answers one question for every role: "what do I do today, and in what order?" The User Manual says what each screen does; this document says when to open it and what "done" looks like.

How to use it.

- Find your role in §4 and follow its Start of day -> During the day -> End of day list. The weekly and monthly lists follow.
- §2 shows how the roles' days fit together, so you can see who is waiting on you.
- §5 is the administrator's data routine: the Excel sync, the lot workbook, backups, updates and Practice.
- §6-§8 are for judgment calls, delays and things that went wrong.
- §9 is one card per role to print and pin up.

Three rules that apply to everyone.

1. Live is real; Practice is for learning. Anything typed in Live moves real stock. To try something, switch to Practice on the sign-in page (violet screens, a pulsing PRACTICE badge - §26 of the manual).
2. Nothing is final until the HOD approves it. A store keeper stages, a supervisor files, the HOD approves. Stock figures update on approval.
3. Fix the source, not the symptom. If a figure is wrong because the Excel workbook is wrong, correct the workbook and sync again; do not "balance" it with an adjustment.

Out of scope: screen-by-screen instructions (User Manual), code and SQL (Architecture), server hosting (manual §17).

§2. The system's day - how the roles fit together

Times are the site's normal working day (Asia/Riyadh, GMT+3). Automatic jobs are marked .

When	Who	What happens	Waiting on it
02:00	System	Nightly database backup (bin/backup_db.sh --install).	Admin checks it weekly (§5.3)
07:00	System	Morning Briefing to admins and HODs - health probes, uncertified Surface Shields, day-shift MTC chase.	HOD, Logistics, QC
07:00-08:00	Store Keeper	Opens the store: incoming deliveries, supervisor requests, overdue returnables, Lots & Expiry red banner.	Supervisors (material), HOD (DNs)
07:00-08:00	Supervisor	Prints the consumption forms for today's jobs; raises material requests.	Store Keeper
Morning	HOD	Clears yesterday's approvals and Delivery Notes.	Store Keeper, Warehouse, Logistics
Morning	Logistics	Incoming PRs -> POs; DN date approvals; overdue deliveries.	Warehouse, HOD
Morning	Warehouse	Incoming assignments, goods received, DNs prepared.	Logistics, HOD
During jobs	Store Keeper	Issues material (FEFO lot), receives deliveries, records returns.	HOD
During jobs	QC	Inspects controlled material; passes, fails or holds it.	Store Keeper (issue is blocked until QC passes)
After each job	Supervisor	Photographs and files the consumption form.	Store Keeper
After each job	Store Keeper	Verifies the supervisor's form; works the Surface Shield queue.	HOD

When	Who	What happens	Waiting on it
Afternoon	HOD	Approves receipts, issues, returns, adjustments, execution entries and Surface Shield jobs.	Everyone who staged something
16:00	System	Evening digest (WhatsApp + bell) and the lot expiry notice - one per site, lots expiring within 30 days or expired.	Store Keeper, HOD
After the workbooks change	Admin	Excel sync (\$5.1), including the Lot Register workbook.	Everyone (stock and lots)
Friday 17:00	System	Weekly executive PDF to admins and HODs.	Management

§3. Role responsibility matrix (RACI)

R = Responsible (does it) - A = Accountable (signs off) - C = Consulted - I = Informed. SK = Store Keeper - Sup = Supervisor - WH = Warehouse User - QC-H = Head of Qualities.

Activity	SK	Sup	HOD	QC	QC-H	Logistics	WH	Admin
Raise a material request for a worker	C	R	I	-	-	-	-	-
Issue stock (with its lot)	R	I	A	-	-	-	-	-
Receive stock at site (batch, MFD, expiry)	R	-	A	C	-	I	-	-
Return stock	R	C	A	-	-	I	-	-
Stock	R	-	A	-	-	-	-	I

Activity	SK	Sup	HOD	QC	QC-H	Logistics	WH	Admin
count / adjustment								
Print and file a consumption form	C	R	A	-	-	-	-	-
Verify a filed consumption form	R	I	A	-	-	-	-	-
Surface Shield job (area, Old/New surface)	R	C	A	-	-	-	-	-
Inspect controlled material	I	-	I	R/A	I	I	I	-
Dispose of / quarantine an expired lot	C	-	A	C	I	-	-	R
Create and submit a PR	-	-	R/A	-	-	I	-	I
Issue a PO, assign a warehouse	-	-	I	-	I	R/A	I	I
Receive from vendor, prepare a DN	-	-	I	C	-	I	R/A	-
Approve	-	-	R/A	-	-	R/A date	I	I

Activity	SK	Sup	HOD	QC	QC-H	Logistics	WH	Admin
a DN (date / content)			content					
Mark a DN received at site	R/A	-	I	-	-	I	I	-
Add / edit an inventory item	-	-	R (own site)	-	-	-	-	R/A (and delete)
WBS numbers and work types	-	-	R/A	-	-	-	-	I
Excel workbook sync (incl. lot workbook)	I	-	C	-	-	-	-	R/A
Keep the workbooks correct	C	-	A	-	-	C	-	R (reports problems)
Users, roles, access requests	-	-	C	-	-	-	-	R/A
Backups, updates, Practice reset	-	-	I	-	-	-	-	R/A
Surface Shield oversight, escalations	-	-	I	C	R/A	I	-	-

§4. SOPs by role

Each role has the same shape: start of day, during the day, end of day, then weekly and monthly, and a short never list. References to the User Manual say manual §; plain § numbers are sections of this SOP.

§4.1 Store Keeper

The store keeper is the only role that moves stock. Everything you stage goes to the HOD.

Start of day (15 minutes)

1. Incoming Deliveries - any DN on its way? Note what to expect today.
2. Supervisor Requests (badge) - approve and issue, or reject with a reason.
3. Lots & Expiry - if the red banner says an expired lot still has stock, check the shelf. If it is there, set it aside and tell the HOD (§6.6). Note the lots in ≤ 30 days: those are what you issue first.
4. Returnable Items - anything overdue? Chase the holder.
5. Read the bell and any What's new notice.

During the day

- Issue Stock - for each issue, pick the material, the tank / work type and (for a Surface Shield) the system code. In the Lot field take the lot marked FEFO unless there is a reason not to; any other lot asks for the reason and the HOD sees it. Never issue the expired lot without the HOD's say-so. For CHEMOLINE, pick the roll. (manual §3.10.3, §4)
- Receive Stock - for a lot-tracked material, type the batch exactly as printed, the MFD and the expiry. If the label has no expiry, leave it blank; GI Hub derives it from the MFD. Attach the DN and, for Surface Shields, the MTC. (manual §3.10.3, §4.4)
- Return Stock - name the lot the material came from. A return gives back to that lot. (manual §3.10.1)
- Mark a DN received when the truck is unloaded and checked (Incoming Deliveries).
- Verify filed consumption forms - Execution Entries -> With store keeper. Compare the supervisor's quantities with what left the shelf. Correct only with a reason; your corrections show to the HOD in red. (manual §4.10.3)
- Surface Shield queue - Execution Entries -> the queue. One card is one job (one tank, one day, one note). Check the area the note filled in, tick the materials that belong to the job, submit to the HOD. (manual §4.9a)

End of day

1. Everything issued today is entered (no "I'll do it tomorrow").
2. Supervisor Requests and With store keeper forms are empty or explained.
3. Anything saved while offline has been sent (the queue indicator is clear).
4. Read the evening expiry notice and plan tomorrow's FEFO issues.

Weekly - Stock Count of the fast movers and every Surface Shield; walk the lot shelves against Lots & Expiry; confirm overdue returnables are chased.

Monthly - full Stock Count; list the lots used but never received (Lots & Expiry, bottom card) and give them to whoever keeps the workbook.

Never - post an adjustment to hide a workbook error; issue from an expired lot without the HOD; re-type a lot number differently from the label (A 4525, not A4525 or 4525); practise in Live.

§4.2 Supervisor

The supervisor runs the field work: asks for the material, records what the job used.

Start of day

1. Execution Entries -> Print a consumption form for each job today. Print one per job; never photocopy a form (its QR code is unique). (manual §4.9)
2. Supervisor -> Material Requests -> New Request for each worker who needs material: worker, job / tank / place, the lines. (manual §5.3)
3. Check My Requests for anything rejected overnight.

During the day

- Fill the paper form on site as the work happens: quantities (packs, or tick KG), area, crew, hours, and the lot on each line.
- Keep the form clean and the QR code uncovered.

End of each job

1. Execution Entries -> Upload a filled form - photograph the whole page, QR code included. Reading takes minutes; you can leave the page. (manual §4.10.1)
2. Check every figure the camera read against the paper, then File.
3. Watch the status: With store keeper -> With HOD -> Approved. A rejection comes back with the reason; correct and re-file.

Weekly - Intent vs Actual: what you asked for against what was issued. Talk to the store keeper about any gap.

Never - photocopy a printed form; file a form that is not yours; ask the store keeper to issue without a request.

§4.3 Head of Department (HOD)

The HOD approves everything the site stages and plans the site's material.

Start of day (30 minutes)

1. Approvals - Receipts, Issues, Returns, Adjustments, Delivery Notes. Clear what was staged yesterday. Read any non-FEFO lot reason and any red store-keeper correction before approving. Reject with a reason, never silently. (manual §6)
2. Execution Entries - forms With HOD: approve, or correct with a written justification (the supervisor is notified).
3. Surface Shield queue -> Awaiting the HOD - High Priority jobs first (more than +/-10% from the benchmark, or no benchmark). Review each job, answer Old or New surface for Garnet, approve or reject the whole job. (manual §4.9a.4, §4.9a.7)
4. Read the Morning Briefing.

During the day

- Lots & Expiry - expired lots with stock: decide dispose / quarantine and ask the Admin to record it

(§6.6). Lots expiring within 30 days: make sure the store keeper is issuing them first.

- Purchase Requests - create and submit PRs to Logistics for what Low Stock and the Material Estimator say is short.
- Cross-Site Requests - request material from another site rather than buying it.
- Records -> Inventory -> New item / Edit - add a material for your site when a new one arrives. (Deleting is the Admin's.)

End of day

1. The approval queues are empty, or every item left has a reason.
2. The evening digest and expiry notice are read.

Weekly - Burn Rate and Low Stock; Lining Coverage; Material Estimator (Session Report, Total Overview) for what can be built now versus when deliveries arrive; Man-Hours variance and the HOD Approval Queue; WBS & Work Types up to date.

Monthly - Executive Summary and Reports -> Monthly Summary for management; Valuation & 30-Day Burn; review the shelf-life and lot list with QC.

Never - approve a whole queue without reading the red and amber values; approve an issue from an expired lot without checking it physically; ask Logistics for a PO directly (raise a PR).

§4.4 Administrator - maintenance and sync

The Admin keeps the data true and the system running. The detailed commands are in §5.

Start of day

1. Admin -> Overdue Actions (red badge) and Access Requests - approve new users into the right role and site.
2. Admin -> Console -> Overview - services, the WhatsApp and email outboxes (retry failures), feedback.
3. If the workbooks were updated yesterday, run the Excel sync (§5.1) and read its report.

During the day

- Users: create, change role or site, reset 2FA, revoke sessions.
- Admin -> Inventory - fix item master data; delete an item only when it has no history.
- Console -> Lots - record a disposal or quarantine the HOD decided (§6.6).
- Triage Feedback (bug reports); copy the prompt for a developer.

End of day - the outboxes are clear; nothing is stuck in Overdue Actions.

Weekly

1. Backups: `./bin/backup_db.sh --list` - last night's file is there (§5.3).
2. Stock vs Excel check - the Stock page's != Excel list; send the causes to whoever keeps the workbook (§5.2).
3. Lots used but never received (Lots & Expiry) - send the list to the workbook keeper.
4. Announcements - tutorial freshness warnings.

Monthly - review users and roles (leavers disabled); restore one backup into a scratch database to prove

it works; Practice reset if trainees have cluttered it (§5.5).

After every update - §5.4.

Never - run a test or a tool against the Live database that is not the documented sync; edit the original workbook with a script (work on a copy); start a second tunnel connector; share a Practice password as a Live one.

§4.5 Logistics

Start of day - Procurement -> Incoming PRs (convert to POs or ask the HOD); DN Approvals (date stage); Purchase Orders - overdue and partial deliveries; Reschedules and Vendor Returns waiting. During the day - Create PO (or Import PO PDF); assign each PO to a warehouse; chase vendors on overdue lines; approve DN dates. End of day - no PR older than its SLA without a note; DN date queue empty. Weekly - open POs review; Lining Coverage for the next month's shortfalls. Monthly - Force-Closures review; vendor performance from Reports. Never - approve DN content (that is the HOD's stage); merge Rubber Lining and Brick Lining material on one DN.

§4.6 Warehouse User

Per shift - Incoming Assignments (acknowledge); receive goods against the PO (batch, MFD and expiry for Surface Shields); Delivery Notes - prepare RL and BL separately, attach the scanned DN before shipping; Returns from Site. Weekly - History review; anything received but not dispatched. Never - ship a DN without its scan; mix RL and BL.

§4.7 Quality Control (QC)

Start of day - Quality -> Inspections: the queue of controlled material waiting. Each item stays blocked from issue until you decide. During the day - pass, fail or hold each one with the certificate checked. A fail mints a return number the store keeper uses. Lots & Expiry: report an expired lot with stock to the HOD. Never - pass material without its MTC; decide outside your site or warehouse.

§4.8 Head of Qualities

Daily - Quality Oversight: Overview, Expired / Expiring / Stagnant, the daily alert. Raise an escalation naming one place for anything that needs a decision. Check Lots & Expiry across sites. Weekly - Surface Shield POs and the MTC Register; Where It Is Used. Monthly - review the 90/60-day thresholds in Settings.

§4.9 Auditor (view-only)

Read Dashboard, Stock, Records, Reports, the HOD read pages and the Material Estimator; download reports. Raise findings through Feedback; the Auditor never changes data.

§5. Data procedures (Administrator)

§5.1 Syncing the Excel workbooks - including the Lot Register workbook

When: after the workbooks have been updated (normally daily), and always after a correction the store or the HOD asked for.

Before: close the workbooks in Excel; put the latest copies in the sync folder. The Lot Register workbook (Rubber & Brick Materials - CNCEC.xlsx, any file named like *Rubber*Brick*CNCEC*.xlsx) goes in the same folder - the sync finds it.

Steps

1. Wake the services if the computer was asleep: `./bin/power.sh wake`.
2. Dry run (shows what would change, changes nothing):

```
cd ~/GI_Hub_Project && DATABASE_URL=postgresql+psycopg2://postgres@127.0.0.1:5433/gihub .venv/bin/python tools/pg_excel_sync.py --site CNCEC --erp --prune-vanished
```

1. Read the report: + new rows, ~ updated rows, - rows removed because the workbook no longer has them, and the > lots section (lots filled, rolls, shelf lives learned, and the problems listed by sheet and row).
2. If it looks right, run the same command with `--commit` at the end.
3. Open Lots & Expiry and the Stock page and spot-check one material.

What the report asks you to fix in the workbook (never in GI Hub):

- a cell with several lots (4525 = 55 Cans; 1823 = 2 Cans) - split the row;
- a lot used but never received - usually a typing slip in the lot or the SAP code; the report names the material that did receive that lot;
- a Lot Register quantity that differs from the Receipt Log, a row with no matching receipt, an expiry before its MFD, a batch spelled differently (0926 vs 0.926);
- a roll typed 10... - corrected to 1O... automatically, but fix the source.

[!] The Lot Register workbook never changes stock. Stock comes only from the Receipt, Consumption and Return Logs. Rows entered in GI Hub are never touched by the sync.

§5.2 Stock vs the Excel workbook

The Stock page marks a material != Excel when GI Hub's stock differs from the workbook's Current Stock, and lists the causes (e.g. Only in GI Hub: an entry made in the app that the workbook does not have). Weekly, send the list to the workbook keeper. To give them a marked copy: `.venv/bin/python tools/stock_excel_check.py --marked` - it writes a copy; never overwrite the original workbook.

§5.3 Backups and restore

- Nightly at 02:00 once installed: `./bin/backup_db.sh --install`.
- Check: `./bin/backup_db.sh --list` - expect last night's file in `.backups/`.
- Before any risky work (a migration, bulk edits): `./bin/backup_db.sh`.
- Restore: `./bin/backup_db.sh --restore FILE` prints the exact command; restore into a scratch database first and compare.

§5.4 After an update

1. Take a backup (\$5.3).
2. Pull the new version and restart (bin/dev.sh or the production service). Database changes apply automatically before anyone signs in.
3. Practice: .venv/bin/python tools/practice_db.py migrate brings both Practice databases to the same version (Practice refuses to start otherwise), then .venv/bin/python tools/practice_db.py verify.
4. If the Live database was reloaded from a dump: re-run tools/practice_db.py wall and backend/scripts/create_ai_readonly_role.sql (a reload wipes both protections).
5. Read the What's new notice yourself so you can answer questions.

§5.5 Practice and training

- New staff learn in Practice with the shared practice.<role> accounts (manual §26.3). Each new Live feature has a Practice example (manual §26.4).
- Reset Practice from Admin -> Console -> Practice (Practice only) when it is cluttered; it returns to the seeded state.
- Training videos and acknowledgements: Training; HODs see compliance.

§5.6 Power - sleep and wake (the office laptop)

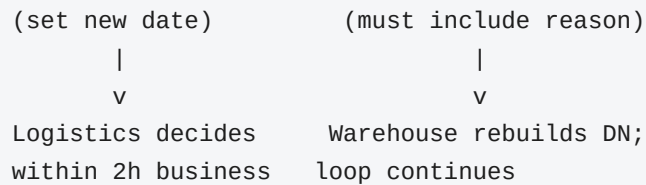
./bin/power.sh sleep stops Postgres and the tunnel to save battery - gi.giiinventory.com is offline while asleep. ./bin/power.sh wake brings both back; ./bin/power.sh status shows what is running.

§6. Decision trees

§6.1-§6.4 are the procurement trees from v1.0; §6.5-§6.7 are new for lots.

§6.1 DN approval - Approve vs Reject vs Reschedule





Approve when: Qty matches PR - Family correct - Lot has > 90d shelf life - Vehicle + driver populated - No HSE concern with this delivery date

Reschedule (don't reject) when: Content is fine but YOU can't receive on the scheduled date (site shutdown, manpower, weather). Reschedule is faster than reject + rebuild.

Reject when: Qty doesn't match PR - Wrong material - Lot expiry inadequate - Wrong destination site (misroute) - Driver/vehicle missing critical info

§6.2 Vendor return vs Stock adjustment

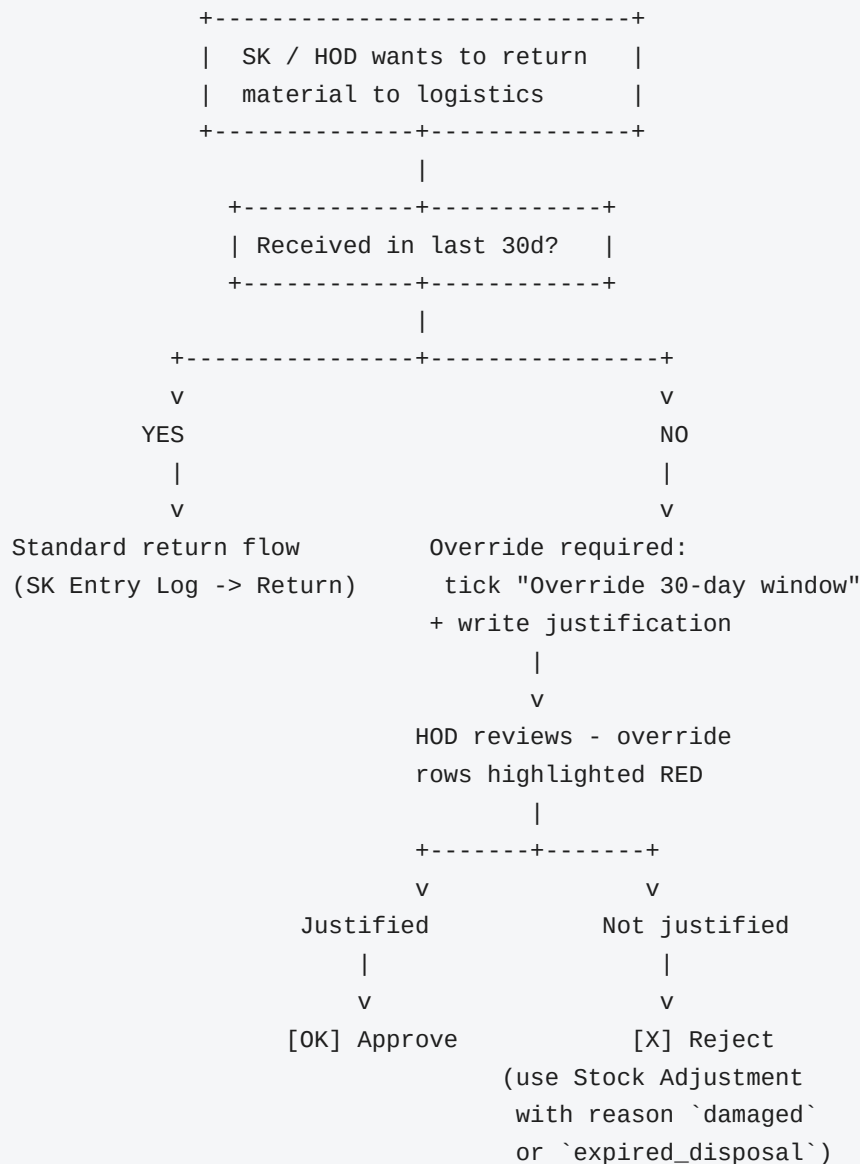


Use Vendor Return when: A specific DN brought defective / wrong qty / quality-issue material. Reopens

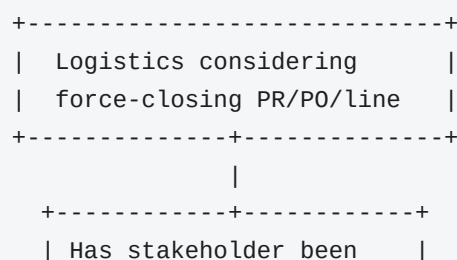
the originating PO line so Logistics can chase vendor.

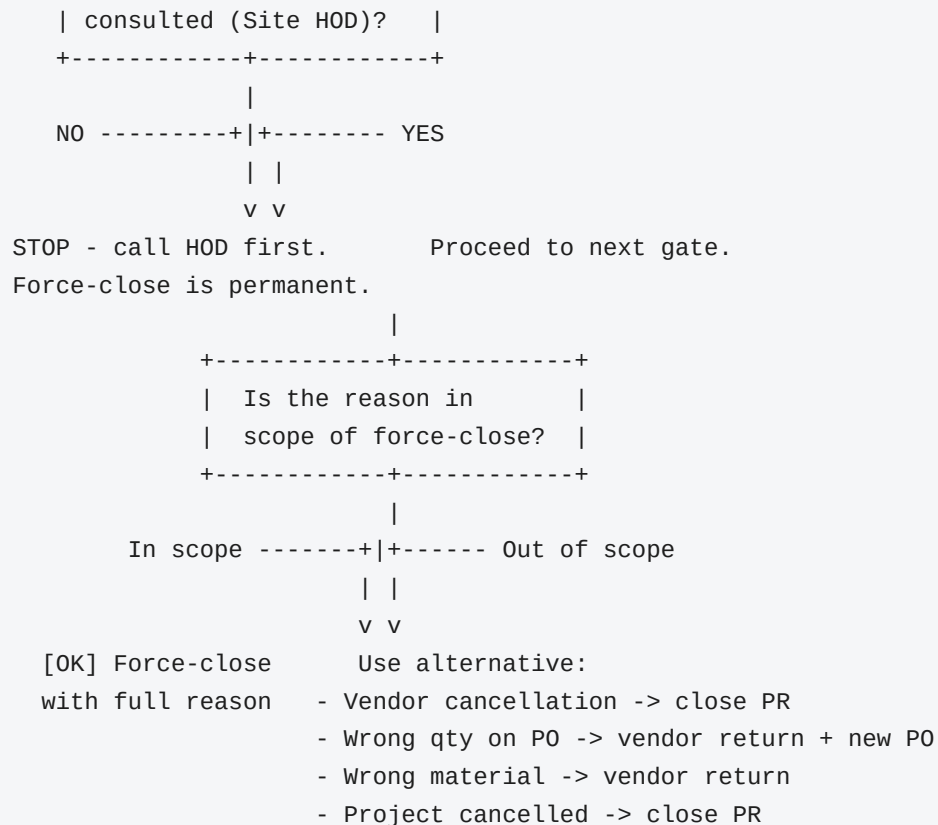
Use Stock Adjustment when: Cycle-count discovers shelf qty != system qty for a general reason (damage in storage, miscount, expired-disposal, etc.). Doesn't touch the PO chain.

§6.3 30-day return window - Override or not?



§6.4 Force-close decision





In scope for force-close: Project cancellation - Vendor permanently unable to fulfill - PR raised in error and beyond email-recall - Item discontinued - Compliance violation

Out of scope (use other tool): Quantity / quality issues (use vendor return) - Date slippage (use reschedule) - Wrong destination site (rebuild DN, no force-close needed)

§6.5 Which lot do I issue?

```

Lot field on Issue Stock (lot-tracked material)
|
+- Take the lot marked FEFO -----> no question asked
|   (earliest valid expiry; lots with no expiry next;
|   expired lots are listed LAST and never suggested)
|
+- A different, valid lot? --> give the reason (e.g. "FEFO lot
|                                   buried behind pallets") -> HOD sees it
|
+- The EXPIRED lot? --> only with the HOD's agreement, after a
|   physical check; give the reason.
|   FEFO warns, it never blocks.

```

A quarantined or disposed lot is not offered at all.

§6.6 An expired lot still has stock

1. Store Keeper: check the shelf.

- It is there -> set it aside, label it, tell the HOD.
- It is not there -> an issue or return was never recorded. Enter it (or ask for the workbook to be corrected). Do not adjust it away.
- 4. HOD: decide with QC - re-test, dispose, or return to the vendor.
- 5. Admin: record the decision in Admin Console -> Lots (Quarantine or Dispose). The lot leaves the Issue form at once.
- 6. If it was disposed, the store keeper posts the matching adjustment with the reason Expired - disposed so stock agrees with the shelf.

§6.7 A lot is "used but never received"

This is a typing problem in a workbook, not a stock problem; the stock already counts the line.

1. Read the row on Lots & Expiry (bottom card): the log, the SAP, the lot as typed, and Received under SAP - the material that did receive a lot with that number.
2. If Received under SAP names a sister item (e.g. lot 3504 consumed under 1043 but received under 1041), the consumption line has the wrong SAP code or the wrong lot. Check the paper form and correct the workbook line.
3. If nothing received that lot anywhere, the lot number itself is mistyped, or the receipt is missing. Check the delivery note.
4. The Admin syncs again (§5.1); the row disappears from the card.

§7. Escalation Matrix

When a process step doesn't happen on time, here's who pings whom, when, and on which channel.

§7.1 PR -> PO escalation

Trigger	Wait time	Action	Channel
PR submitted to Logistics, no PO after 4 business hours	4h	Site HOD -> Logistics lead	WhatsApp
^ Still nothing after another 4h	8h	Site HOD -> Admin	App notification + email
^ Still nothing after 24h	24h	Admin overrides - assigns alternate Logistics user	Admin Portal -> Users

§7.2 PO -> Assignment escalation

Trigger	Wait time	Action	Channel
PO issued, no warehouse assignment after 8 business hours	8h	Site HOD -> Logistics (with PO number)	WhatsApp

Trigger	Wait time	Action	Channel
Assignment created, Warehouse hasn't acknowledged in 24h	24h	Logistics -> Warehouse Lead	WhatsApp + email
^ Still no ack after 48h	48h	Warehouse Lead -> Admin	Admin escalation

§7.3 DN approval escalation

Trigger	Wait time	Action	Channel
DN sitting at pending_logistics for > 4 business hours	4h	Warehouse -> Logistics	WhatsApp
DN sitting at pending_hod for > 8 business hours	8h	Logistics -> Site HOD	WhatsApp
DN sitting at pending_sk for > 24 hours after expected arrival	24h	HOD -> SK + Warehouse	WhatsApp

§7.4 Reschedule escalation

Trigger	Wait time	Action	Channel
Reschedule pending > 2 business hours during work day	2h	Requester -> Logistics	WhatsApp
^ Pending > 24h	24h	Site HOD -> Admin	App + email

§7.5 Vendor return escalation

Trigger	Wait time	Action	Channel
Vendor return raised, no acknowledgement from vendor in 7 days	7d	Logistics -> Vendor directly (email)	Email
^ Still unresolved after 14d	14d	Logistics -> Site HOD + Admin	App + email
Resupply date passed without delivery	T+1d	Logistics -> Admin	Critical

§7.6 Force-closure escalation

Force-closures fire critical notifications to Admin + originating Site HOD instantly. There's no escalation needed - the action is already at maximum severity. Admin reviews weekly per §3.13.

§8. Recovery procedures

§8.1-§8.7 are the procurement procedures from v1.0; §8.8-§8.10 are new.

§8.1 DN rejected at HOD - how Warehouse recovers

1. Bell shows rejection notification with HOD's reason
2. Warehouse user reads the reason from the notification card (full text in the related DN row in delivery_notes.rejection_reason)
3. Determine what changed:
 - If qty wrong: prepare a new DN with corrected qty (the rejected DN's qty frees back into the available calculation immediately)
 - If material wrong: file an internal Stock Adjustment if you over-counted at receive, OR raise a vendor return if the wrong material was delivered to you
 - If date wrong: prepare new DN with the corrected date; HOD will approve in same cycle
7. Submit new DN -> Logistics -> HOD; loop continues

§8.2 Warehouse short-receives from vendor

1. Receive Goods tab - type the actual received qty (less than ordered)
2. System auto-flips po_items.line_status to partially_delivered
3. Add a note in the assignment notes describing the shortfall
4. Notify Logistics via WhatsApp with the PO number + shortfall qty
5. Logistics raises a Vendor Return with reason "Vendor under-shipped" (uses the same return flow, reopens the line for resupply)
6. New shipment from vendor -> Warehouse receives the remainder -> line flips to delivered

§8.3 Vendor return + resupply cycle

1. Issue surfaces (SK at site OR Warehouse at receive)
2. Determine raiser:
 - Issue at site after SK confirmed -> SK raises to HOD -> HOD raises to Warehouse via internal return -> Warehouse raises Vendor Return
 - Issue at warehouse before DN ship -> Warehouse raises Vendor Return directly
5. Logistics tracks return in <- Vendor Returns tab
6. Expected_Resupply date set (default today + 14)
7. T-2 / T-1 / T-0 reminders fire on Expected_Resupply
8. When resupply arrives: Warehouse re-receives -> line Delivered_Qty recalculates -> status moves forward

§8.4 HOD absent for > 24h with pending DN

1. Admin shadows HOD Portal (admin can access via the existing override)
2. Admin reviews the pending DN(s) in HOD Portal -> DN Approvals
3. Admin approves on behalf, leaves a note in the DN attesting to the cover

4. Audit log records DN_HOD_APPROVE with admin's username
5. HOD-on-return is notified to review post-fact

§8.5 Logistics absent - Admin override

1. Admin enters Logistics Portal via shadow access
2. Admin acts as Logistics for the day (approve PRs, issue POs if pre-arranged, approve reschedules)
3. Audit log records each action with admin's username (not the absent Logistics user)
4. Admin notifies the absent Logistics user via WhatsApp before signing off

§8.6 Force-closure made in error

There's no "undo force-close" button (yet - on the v3.0 backlog). Workaround:

1. Admin -> Master DB Editor -> relevant table (purchase_orders / pr_master)
2. Edit status from force_closed back to open (PO) or remove the logistics_status='force_closed' flag (PR)
3. Add a corresponding po_force_closures row with reason "Reversed in error - see audit"
4. The Site HOD's "Force-closures affecting me" tab will show both the original close + the reversal

Use sparingly - this is invasive. Better: communicate the reversal verbally and update via DB Editor with a paper trail.

§8.7 RL/BL DN was prepared as combined (system rejected, user confused)

1. UI shows: "Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."
2. Warehouse user reduces Ship Qty to 0 on either the RL lines OR the BL lines (not both)
3. Save DN draft - succeeds with only one family
4. Prepare second DN with the other family - same source PO, different family
5. Submit both DNs separately; both flow through approval independently

§8.8 The Excel sync reports problems

The sync never stops for a workbook problem; it reports it and carries on with the rows it can trust. Work the report top to bottom:

1. Rows it could not place (unknown SAP, unreadable date or quantity) - fix the workbook cell; nothing was imported for that row.
2. Multi-lot cells - split the row, one lot per row.
3. Lots used but never received - §6.7.
4. Lot Register differences (quantity vs Receipt Log, no matching receipt, expiry before MFD) - correct whichever workbook is wrong. Stock is not affected either way.
5. Sync again and confirm the list is shorter.

If a sync was committed by mistake, restore the backup taken before it (§5.3) - and from then on take one before every sync.

§8.9 An entry was saved while the signal was down

The app keeps it on the device and sends it when the connection returns; the queue indicator shows how many are waiting. It is sent once, however many times it retries. Do not type it again. If the device is lost or reset before it reconnects, the entry is lost - enter it again from the paper.

§8.10 Something was entered in Practice instead of Live (or the other way)

- Practice by mistake: nothing reached Live. Enter it again in Live. Practice and Live are separate systems, and Practice entries never move to Live - not even entries saved offline.
- Live by mistake (a training entry): ask the HOD to reject it if it is still pending. If it was approved, the store keeper reverses it with an adjustment whose reason says what happened, and the HOD approves that.

§9. Quick reference cards

§9.1 Store Keeper

- Morning: Incoming Deliveries - Supervisor Requests - Lots & Expiry banner - Returnables.
- Issue: take the FEFO lot; any other lot needs a reason; never the expired lot without the HOD.
- Receive: batch as printed - MFD - expiry (blank = derived) - DN + MTC.
- Verify supervisor forms; submit Surface Shield jobs (one card = one job).
- Evening: nothing left unentered; read the expiry notice.

§9.2 Supervisor

- Print one form per job (never photocopy).
- Material Requests -> New Request for each worker.
- After the job: photograph the whole form (QR in view) -> check -> File.
- Watch: With store keeper -> With HOD -> Approved.

§9.3 HOD

- Morning: Approvals (all five tabs) - forms With HOD - Surface Shield jobs (High Priority first; Old/New for Garnet).
- Read every red / amber value and every non-FEFO reason before approving.
- Expired lot with stock -> decide with QC -> Admin records it.
- PRs for what is short; Cross-Site before buying.
- Weekly: Burn Rate, Low Stock, Lining Coverage, Estimator, Man-Hours.

§9.4 Administrator

- Daily: Overdue Actions - Access Requests - Console outboxes - Excel sync (dry run -> read -> --commit).
- Weekly: backup list - Stock vs Excel - lots used but never received - tutorial freshness.

- After an update: backup -> restart -> practice_db.py migrate + verify.
- Never run tools against Live except the documented sync; never edit the original workbook with a script.

§9.5 Logistics

- Incoming PRs -> POs -> assign warehouse - DN date approvals - overdue lines - reschedules - vendor returns.

§9.6 Warehouse User

- Acknowledge assignments - receive against PO - DNs RL and BL separate, scan attached - returns from site.

§9.7 QC - Head of Qualities - Auditor

- QC: Inspections queue -> pass / fail / hold with the MTC; report expired lots.
- Head of Qualities: Quality Oversight daily; escalate naming one place.
- Auditor: read and download only; findings through Feedback.

§10. Glossary

For the full vocabulary see USER_MANUAL.md §11.5. Terms used in this SOP:

Term	Meaning
Live / Practice	Live is the real system. Practice is a separate system with invented data for learning (violet screens, PRACTICE badge). Nothing crosses between them.
Stage / approve	A store keeper or supervisor stages an entry; it changes stock only when the HOD approves it.
Lot (batch)	One batch of a material, e.g. PU COMP A batch 3504. Lot-tracked materials are the Surface Shields except bricks.
FEFO	First Expire, First Out - issue the lot that expires soonest. In GI Hub expired lots go last and FEFO warns, never blocks.
MFD	Manufacture date. A missing expiry is MFD + the item's shelf life (derived).
Roll register	CHEMOLINE rolls, each a unit inside its batch (10..., letter O).
Lot Register workbook	Rubber & Brick Materials - CNCEC.xlsx - adds MFD, expiry and rolls to lots; never changes stock.
Used but never received	A lot named on a consumption or return line that no

Term	Meaning
	receipt of that material brought in - a workbook typing slip.
Quarantined / Disposed	A lot the Admin took out of use (Admin Console -> Lots). Never offered on the Issue form.
Excel sync	tools/pg_excel_sync.py - makes GI Hub's workbook-sourced rows match the workbooks (adds, updates, removes); never touches rows entered in GI Hub.
Job (Surface Shield)	One tank on one day for one store-keeper note; its area is credited once.
Old / New surface	The HOD's answer for a Garnet (blasting) job; it decides which benchmark the draw is compared with.
Procurement chain	Site PR -> Logistics PO -> Warehouse DN -> Site receipt, all in the app.
DN state machine	draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received; rejected is terminal from any pending state.
RL/BL strict separation	Rubber Lining and Brick Lining never share a PO group, DN or warehouse aggregation.
Warehouse-blind pricing	A warehouse user never sees prices on a PO.
Shadow (Logistics / Warehouse)	The Admin's access to those portals for absence cover or audit.
Force-closure	Logistics' one-way close of a PR / PO / line with a mandatory reason; notifies Admin and the site HOD.
T-2 / T-1 / T-0 reminders	Daily notices 2 days before, 1 day before and on the expected delivery date.
Material Estimator (SME)	HOD planning portal for lining material. It has its own material pool (from the estimator workbook), separate from the store's stock; available is what has arrived, ordered is the total procured.
Man-Hours	HOD labour tracking: roster, timesheets, estimator, variance. Writes only mh_* tables.
OWN vs Supply	Own (GI) staff vs subcontractor labour.

§11. Change Log

Version	Date	Author	Changes
1.0	2026-06	Initial release	Procurement chain: Site

Version	Date	Author	Changes
			HOD, Logistics, Warehouse User, SK, Admin. RACI, cadences, 4 decision trees, escalation matrix, 7 recovery procedures, 5 quick cards.
1.1	2026-06-28	Update	Man-Hours and Material Estimator HOD cadences; glossary entries.
2.0	2026-10-03	Rewrite	Whole system, by role, for the React / FastAPI application. New: the system's day (§2); RACI for all nine roles (§3); daily plans for Store Keeper, Supervisor, HOD, Admin, Logistics, Warehouse, QC, Head of Qualities, Auditor (§4); the administrator's data procedures - Excel and Lot Register sync, stock vs Excel, backups, updates, Practice, power (§5); decision trees for lots and FEFO (§6.5-6.7); recovery for sync problems, offline entries and Practice/Live mistakes (§8.8-8.10); new quick cards (§9); lot and Practice terms in the glossary. Procurement trees, escalation and recovery from v1.0 kept as §6.1-6.4, §7 and §8.1-8.7.

End of SOP. Review quarterly. Companion documents: USER_MANUAL.md (every screen) - MANUAL_TESTING_GUIDE.md (how each feature is checked) - docs/ARCHITECTURE.md (how it is built).